

DATA ANALYST CAPSTONE PROJECT

Mutual Fund Analytics

Python | SQL | Power BI | Financial Analytics

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Project Overview



Business Problem

The need for a centralized, data-driven approach to evaluate mutual fund performance, track market trends, and decode underlying investor transaction patterns.



Project Objectives

Consolidate disparate data sources, engineer robust financial metrics, and build an interactive dashboard to empower strategic investment decisions.



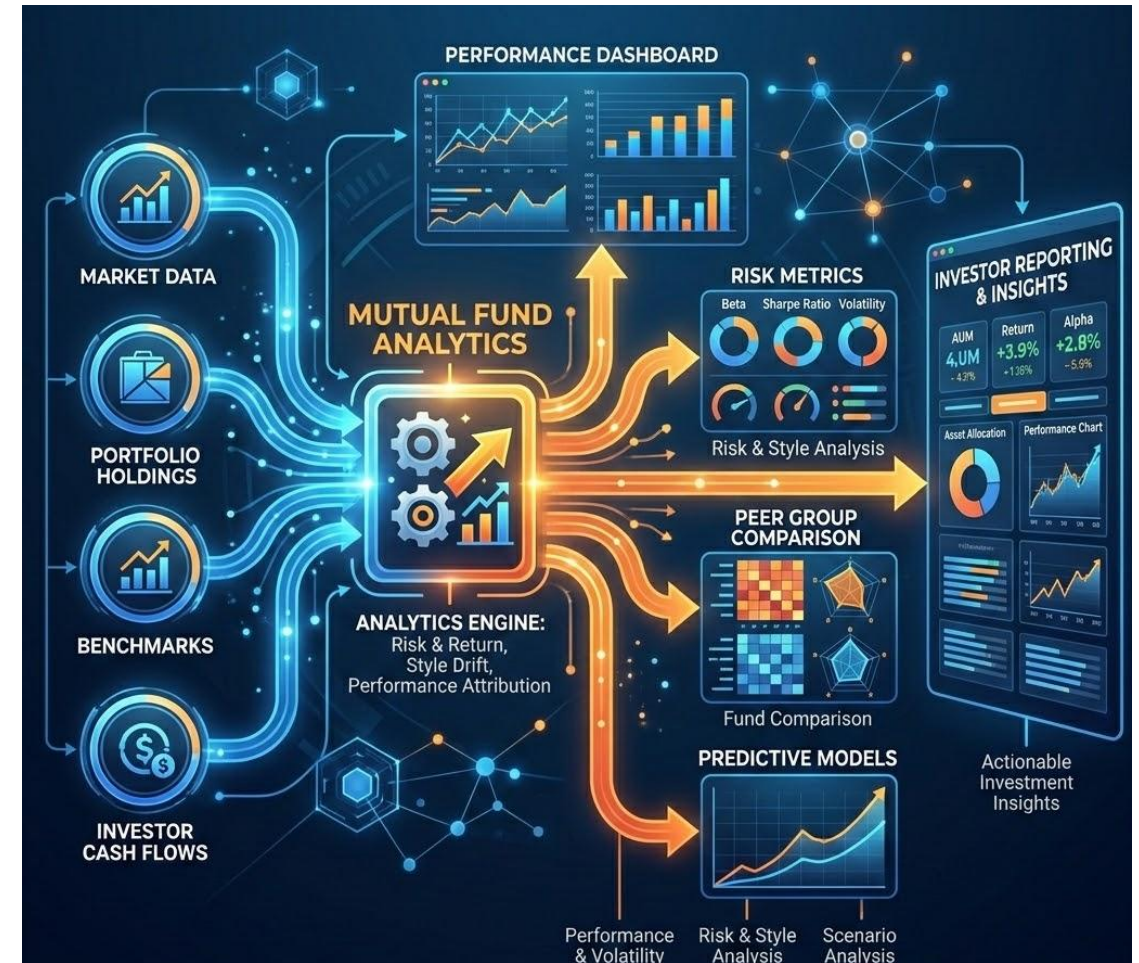
Expected Outcomes

A comprehensive BI dashboard providing actionable insights on fund risk-adjusted returns, SIP inflow trends, and target demographic behaviors.

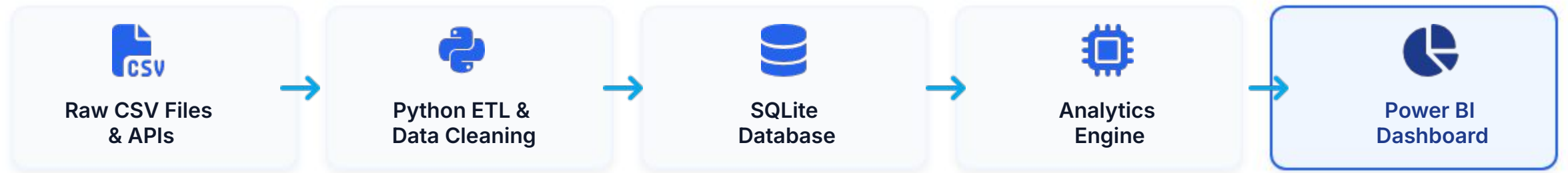
Comprehensive Data Sources

Total Sources Used: 10 Raw Datasets + MFAPI +
Generated Analytics Tables

-  **Fund Master Data:** AMC details, fund categories, launch dates.
-  **NAV History:** Historical pricing data via direct MFAPI integration.
-  **AUM Data:** Asset Under Management metrics by fund house.
-  **SIP Inflows:** Monthly systematic investment tracking.
-  **Investor Transactions:** Granular purchase/redemption logs.
-  **Portfolio Holdings:** Asset allocation and sector breakdown.
-  **Benchmark Data:** NIFTY 50 and index comparative baselines.



Data Pipeline & Architecture



1. Extraction

Gathering multiple mutual fund datasets from CSV files and APIs.

2. Transformation

Handling missing values, standardizing dates.

3. Load

Relational mapping using SQLAlchemy.

4. Visualization

Interactive DAX measures & reporting.

Technology Stack



Python

Core
Scripting



Pandas & NumPy

Data
Manipulation



SQLite

Data Storage



SQLAlchemy

ORM Integration



Power BI

Data
Visualization



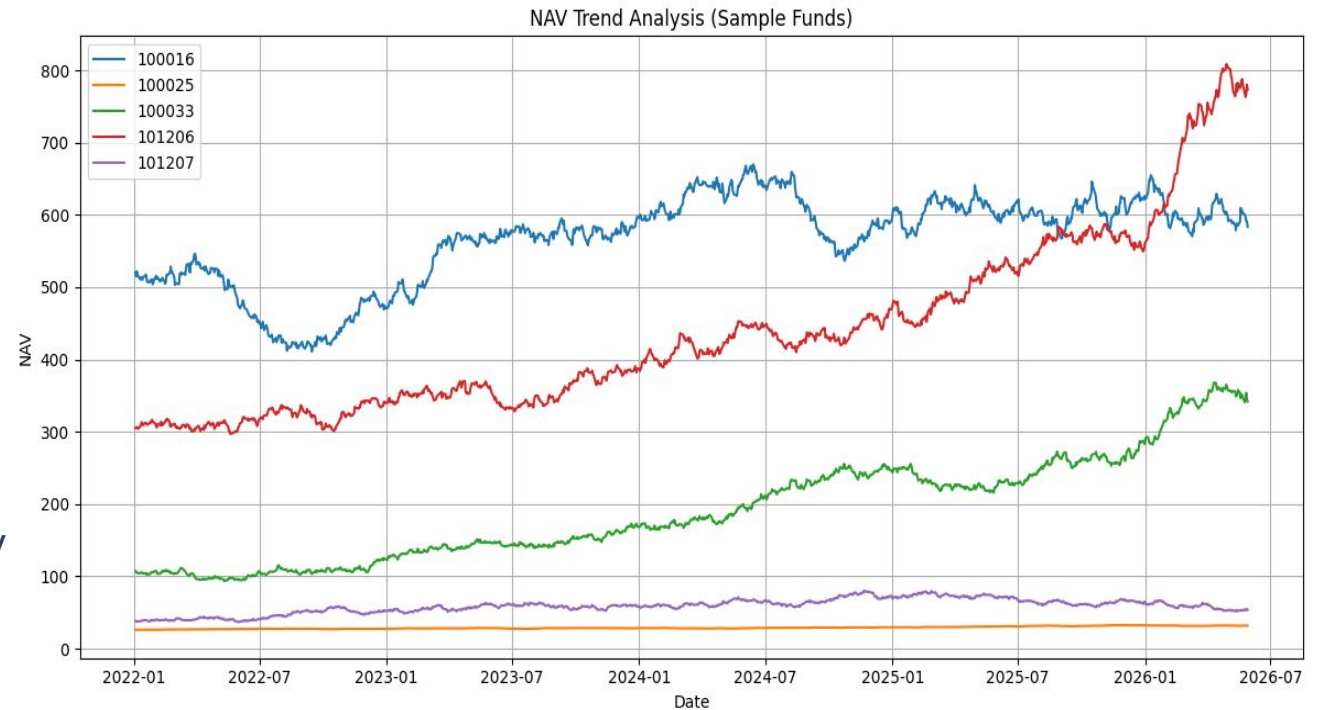
GitHub

Version
Control

Exploratory Data Analysis (EDA)

Key Market Insights

- ✓ NAV values showed a consistent upward trend across most equity-oriented funds.
- ✓ Monthly SIP inflows increased steadily throughout the analysis period.
- ✓ Industry AUM demonstrated strong long-term growth.
- ✓ Investor transactions were concentrated across key age groups and major states.
- ✓ Correlation analysis revealed varying relationships between fund categories.



NAV Trend Analysis (2022-2026)

Performance Analytics & Risk Metrics

Key Metrics Calculated:

- CAGR (Compound Annual Growth Rate)
- Sharpe Ratio (Risk-Adjusted Return)
- Sortino Ratio (Downside Risk-Adjusted Return)
- Alpha & Beta Analysis
- Maximum Drawdown
- Fund Scorecard Ranking

Project Highlights:

- Best Fund Score Achieved: 100
- Top Ranked Fund: AMFI 120505
- Highest Observed Sharpe Ratio: 1.09
- Highest Observed Sortino Ratio: 1.83
- Maximum Drawdown Range: -4.31% to -35.45%

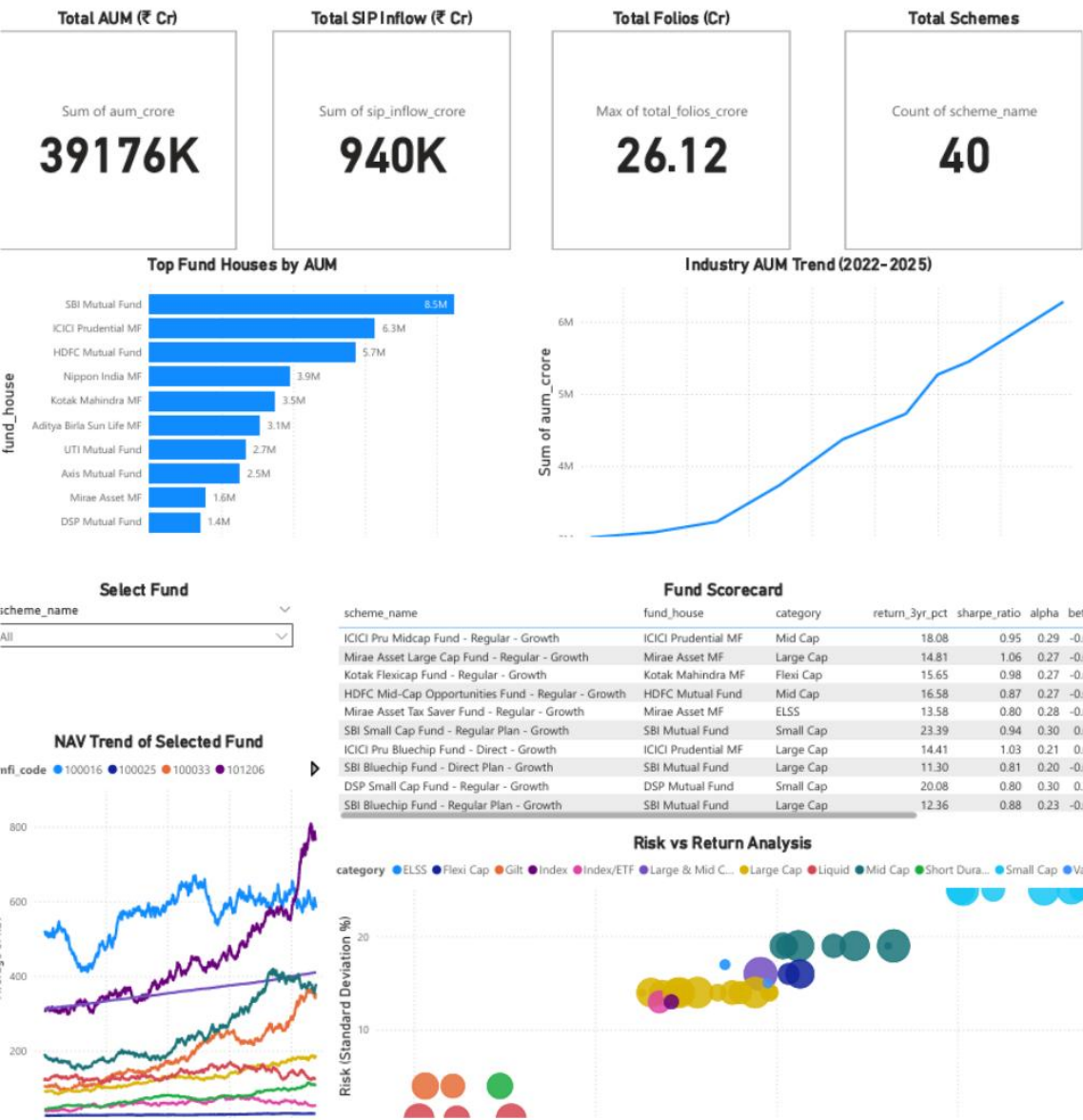
Metric	Value
Highest Sharpe Ratio	1.09
Highest Sortino Ratio	1.83
Best Fund Score	100
Lowest Drawdown	-4.31%

BI Dashboard Overview

The interactive Power BI dashboard serves as the central hub for our insights, seamlessly connecting the backend SQL database to front-end visual analytics.

Primary KPIs Tracked

- **Total AUM:** Industry-wide asset growth tracking
- **Total SIP Inflow:** Monthly retail investment monitoring
- **Fund Performance:** NAV trend and scorecard analysis.
- **Risk vs Return:** Risk-adjusted fund comparison



Select Fund

scheme_name

All

Fund Scorecard

scheme_name	fund_house	category	return_3yr_pct	sharpe_ratio	alpha	beta
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	Mid Cap	18.08	0.95	0.29	-0.02
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Large Cap	14.81	1.06	0.27	-0.02
Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	Flexi Cap	15.65	0.98	0.27	-0.02
HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	Mid Cap	16.58	0.87	0.27	-0.02
Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	ELSS	13.58	0.80	0.28	-0.02
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Small Cap	23.39	0.94	0.30	0.02
ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	Large Cap	14.41	1.03	0.21	0.02
SBI Bluechip Fund - Direct Plan - Growth	SBI Mutual Fund	Large Cap	11.30	0.81	0.20	-0.02
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	Small Cap	20.08	0.80	0.30	0.12
SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	Large Cap	12.36	0.88	0.23	-0.04

NAV Trend of Selected Fund

nfi_code

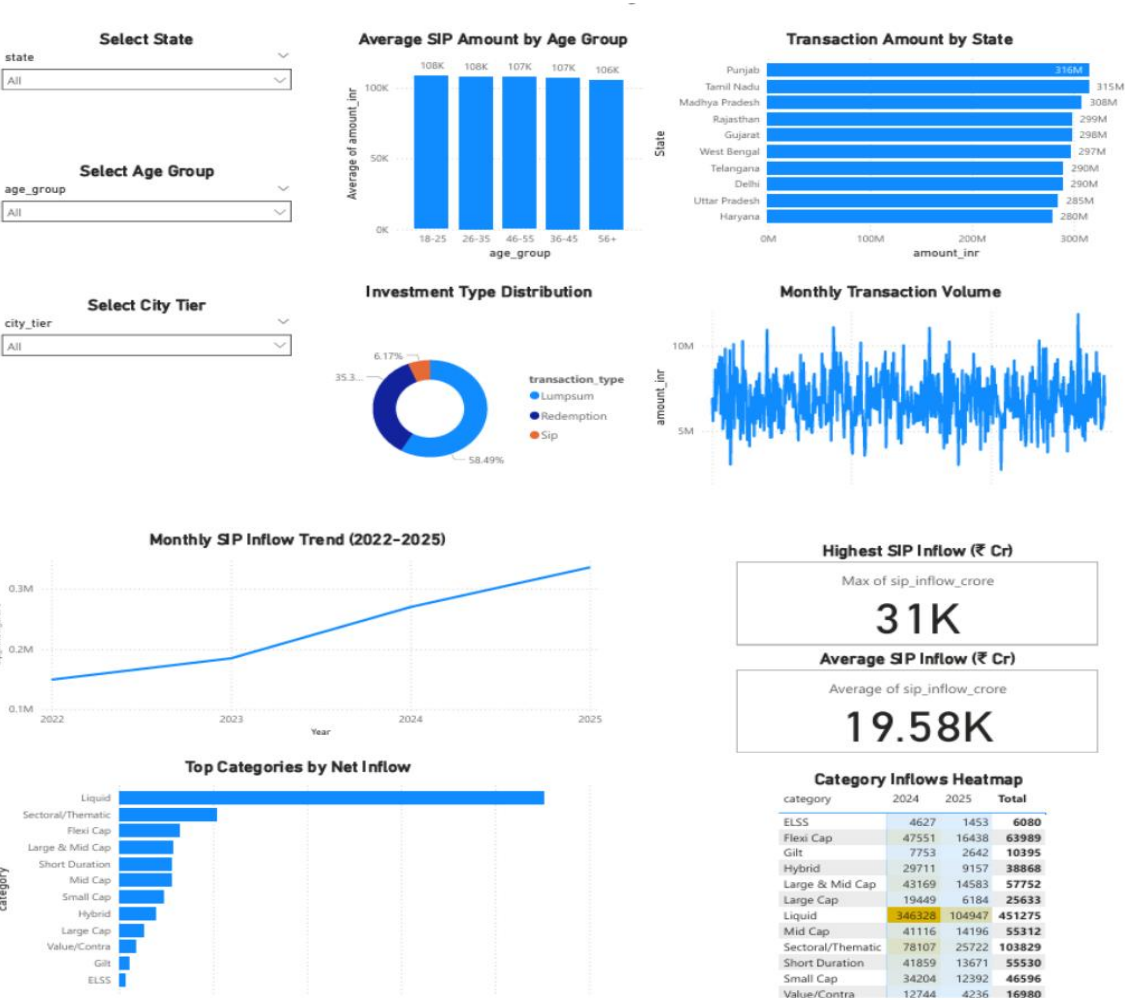
100016 100025 100033 101206

Risk vs Return Analysis

category

ELSS Flexi Cap Gilt Index Index/ETF Large & Mid C... Large Cap Liquid Mid Cap Short Dura... Small Cap Vali

Investor Analytics Dashboard



Key Investor Insights

- Investor activity was analyzed across age groups and city tiers.
- State-wise transaction patterns highlighted regional investment concentration.
- SIP transactions represented the dominant recurring investment mode.
- Monthly transaction volume showed consistent participation throughout the analysis period.

Advanced Analytics & Modeling

VaR & CVaR

Modeled downside risk using historical fund returns.

Rolling Sharpe Ratio

Tracked rolling Sharpe ratios to evaluate consistency of risk-adjusted performance over time.

Cohort Analysis

Grouped investors by onboarding month to track long-term portfolio growth and retention behavior.

SIP Continuity

Identified investors at risk of SIP discontinuation using transaction frequency and average gap analysis.

Sector HHI Analysis

Utilized Herfindahl-Hirschman Index to measure portfolio concentration and sector diversification risks.

Recommendation Engine

Built logic to map investor risk profiles to suitable mutual fund recommendations using historical fund performance data.

Key Findings & Strategic Recommendations



Key Findings

Sustained AUM Growth: The industry shows robust compounded growth despite short-term volatility.

SIP Dominance: SIP transactions represented the dominant recurring investment mode across investors.

Risk Adjustments Matter: Several high-return funds failed on risk metrics (high Max Drawdown).

Diversification pays off: Lower Sector HHI values indicated better portfolio diversification and reduced concentration risk.



Targeted Recommendations

Data-driven portfolio alignment based on investor profiles:

**Low Risk
Appetite**

→ Liquid & Ultra-Short Debt Funds

Moderate Risk Appetite
Advantage Funds

→ Large Cap & Balanced

**High Risk
Appetite**

→ Mid/Small Cap & Sectoral Funds

Thank You!

Transforming Mutual Fund Data into Actionable Investment Insights

Core Skills Demonstrated:

Python

SQL

Power BI

Data Analytics

Financial Modeling

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 <https://github.com/sumit123-tech/bluestock-mf-capstone>